## Analytics

**Amazon Athena**

- **What it is:**
  - Serverless query service to run SQL directly on data stored in S3
- **Core Features:**
  - No infrastructure (fully serverless)
  - Uses Presto/Trino engine
  - Supports CSV, JSON, Parquet, ORC
  - Integrated with AWS Glue Data Catalog
- **When to Use:**
  - Ad-hoc data analysis
  - Log analysis (CloudTrail, ALB logs)
  - Querying large datasets in S3
- **Not Ideal For:**
  - High-frequency queries
  - Low-latency applications
- **Exam Tips :**
  - Pricing = per data scanned
  - Use partitioning + columnar formats (Parquet)
  - Often paired with S3 + Glue

**Amazon EMR**

- **What it is:**
  - Managed big data platform for Hadoop, Spark, Hive workloads
- **Core Features:**
  - Supports Hadoop ecosystem
  - Cluster-based processing
  - Integrates with S3

- Supports Spot instances

• **When to Use:**

- Large-scale batch processing
- Big data analytics

• **Not Ideal For:**

- Real-time processing
- Serverless workloads

• **Exam Tips :**

- EMR = big data + cluster
- Use Spot instances for cost savings

**AWS Glue**

• **What it is:**

- Serverless ETL service

• **Core Features:**

- Glue Data Catalog
- Schema discovery via crawlers
- PySpark support
- Serverless execution

• **When to Use:**

- Data transformation pipelines
- Preparing data for analytics

• **Not Ideal For:**

- Real-time streaming
- Simple queries

• **Exam Tips :**

- Glue = ETL + Catalog
- Works with Athena + S3

**AWS Glue DataBrew**

- **What it is:**
  - No-code data preparation tool
- **Core Features:**
  - Visual interface
  - Pre-built transformations
  - Data profiling
- **When to Use:**
  - Data cleaning
  - Preparing datasets
- **Not Ideal For:**
  - Complex ETL pipelines
- **Exam Tips :**
  - DataBrew = no-code data cleaning

**AWS Lake Formation**

- **What it is:**
  - Service to build and manage data lakes
- **Core Features:**
  - Centralized permissions
  - Data catalog integration
  - Fine-grained access control
- **When to Use:**
  - Data lake management
  - Governance
- **Not Ideal For:**
  - Data processing
- **Exam Tips :**
  - Lake Formation = data lake + security

**Amazon Kinesis Data Firehose**

• **What it is:**

- Fully managed service to deliver streaming data

• **Core Features:**

- No shard management
- Auto scaling
- Direct delivery to S3, Redshift, OpenSearch

• **When to Use:**

- Streaming → storage pipelines
- Log ingestion

• **Not Ideal For:**

- Custom processing

• **Exam Tips :**

- Firehose = delivery (no management)

**Amazon Kinesis Data Streams**

• **What it is:**

- Real-time streaming data ingestion service

• **Core Features:**

- Shard-based scaling
- Data retention
- Ordered data per shard

• **When to Use:**

- Real-time analytics
- Event streaming

• **Not Ideal For:**

- Simple delivery

• **Exam Tips :**

- Shard = throughput
- Streams = custom processing

**Amazon Managed Service for Apache Flink**

• **What it is:**

- Managed service for real-time stream processing

• **Core Features:**

- Stateful processing
- Low latency
- Integrates with Kinesis

• **When to Use:**

- Complex real-time analytics

• **Not Ideal For:**

- Batch workloads

• **Exam Tips :**

- Flink = real-time processing engine

**Amazon Managed Streaming for Apache Kafka**

• **What it is:**

- Managed Apache Kafka service

• **Core Features:**

- Fully managed Kafka clusters
- High throughput
- Kafka API compatibility

• **When to Use:**

- Kafka-based systems

• **Not Ideal For:**

- Simple streaming

• **Exam Tips :**

- MSK = Kafka on AWS

**Amazon OpenSearch Service**

• **What it is:**

- Managed search and log analytics service

• **Core Features:**

- Full-text search
- Dashboards (Kibana/OpenSearch)
- Near real-time indexing

• **When to Use:**

- Log analytics
- Search features

• **Not Ideal For:**

- Transactional databases

• **Exam Tips :**

- Kinesis → OpenSearch pipeline common

**Amazon QuickSight**

• **What it is:**

- Business Intelligence (BI) dashboard service

• **Core Features:**

- Interactive dashboards
- SPICE in-memory engine
- Serverless scaling

• **When to Use:**

- Reporting
- Data visualization

• **Not Ideal For:**

- Data processing

- **Exam Tips :**
  - QuickSight = dashboards + BI

**Amazon SageMaker AI**

- **What it is:**
  - Fully managed platform for machine learning
- **Core Features:**
  - Model building, training, deployment
  - Managed notebooks
  - AutoML
  - Real-time inference
- **When to Use:**
  - AI/ML applications
  - Predictive analytics
- **Not Ideal For:**
  - Simple analytics
- **Exam Tips :**
  - SageMaker = full ML lifecycle
  - Build → Train → Deploy

## Application Integration

**Amazon AppFlow**

- **What it is:**
  - Fully managed service to transfer data between SaaS applications and AWS
- **Core Features:**
  - No-code data integration
  - Supports Salesforce, SAP, Google Analytics, etc.
  - Scheduled or event-based data flows
  - Data transformation capabilities

- **When to Use:**
  - SaaS data integration
  - Moving data into S3, Redshift
- **Not Ideal For:**
  - Complex ETL pipelines (use Glue)
- **Exam Tips:**
  - AppFlow = SaaS data transfer (no code)

**Amazon EventBridge**

- **What it is:**
  - Serverless event bus for routing events between AWS services
- **Core Features:**
  - Rule-based event filtering
  - Supports AWS + SaaS integrations
  - Multiple targets (Lambda, SQS, Step Functions)
- **When to Use:**
  - Event-driven architectures
  - Microservices communication
- **Not Ideal For:**
  - Message buffering (use SQS)
- **Exam Tips:**
  - EventBridge = event routing + filtering

**Amazon Managed Workflows for Apache Airflow**

- **What it is:**
  - Managed Apache Airflow service for workflow orchestration
- **Core Features:**
  - DAG-based workflows
  - Managed Airflow environment

- Scheduling and monitoring

• **When to Use:**

- Data pipelines
- ETL orchestration

• **Not Ideal For:**

- Simple workflows (use Step Functions)

• **Exam Tips:**

- MWAA = Airflow + DAG pipelines

**Amazon Simple Notification Service**

• **What it is:**

- Managed publish/subscribe (push-based) messaging service

• **Core Features:**

- Push notifications (email, SMS, Lambda, HTTP)
- Fan-out architecture
- Topic-based messaging

• **When to Use:**

- Alerts and notifications
- Broadcasting messages

• **Not Ideal For:**

- Message storage (use SQS)

• **Exam Tips:**

- SNS = push model + fan-out

**Amazon Simple Queue Service**

• **What it is:**

- Managed message queue (pull-based)

• **Core Features:**

- Standard and FIFO queues

- Dead Letter Queue (DLQ)
- Message retention

• **When to Use:**

- Decoupling applications
- Asynchronous processing

• **Not Ideal For:**

- Real-time push notifications

• **Exam Tips:**

- SQS = decoupling + buffering

**AWS Step Functions**

• **What it is:**

- Serverless workflow orchestration service

• **Core Features:**

- State machine workflows
- Built-in retries and error handling
- Parallel execution

• **When to Use:**

- Multi-step workflows
- Microservices orchestration

• **Not Ideal For:**

- Complex DAG pipelines (use MWAA)

• **Exam Tips:**

- Step Functions = workflow + retries

## Cloud Financial Management

**AWS Budgets**

• **What it is:**

- Service to set budgets and receive alerts

• **Core Features:**

- Cost and usage budgets
- Alerts via email/SNS
- Forecasting

• **When to Use:**

- Cost control
- Budget tracking

• **Not Ideal For:**

- Detailed cost analysis

• **Exam Tips:**

- Budgets = alerts on cost thresholds

**AWS Cost Explorer**

• **What it is:**

- Tool to analyze and visualize AWS costs

• **Core Features:**

- Cost breakdown by service
- Historical trends
- Forecasting

• **When to Use:**

- Cost analysis
- Usage insights

• **Not Ideal For:**

- Real-time alerts (use Budgets)

• **Exam Tips:**

- Cost Explorer = cost visualization

## Compute

**AWS Batch**

- **What it is:**
  - Fully managed service to run **batch computing jobs** at scale
- **Core Features:**
  - Automatically provisions compute resources
  - Supports EC2 and Fargate
  - Job queues and scheduling
  - Scales based on workload
- **When to Use:**
  - Large batch processing jobs
  - Data processing, simulations
- **Not Ideal For:**
  - Real-time processing
  - Event-driven applications
- **Exam Tips:**
  - Batch = **large-scale batch jobs (no infra management)**

**Amazon EC2**

- **What it is:**
  - Virtual machines in the cloud
- **Core Features:**
  - Full OS control
  - Multiple instance types
  - Auto Scaling integration
  - Flexible pricing (On-Demand, Reserved, Spot)
- **When to Use:**
  - Custom applications
  - Long-running workloads
- **Not Ideal For:**
  - Serverless/event-driven workloads

- **Exam Tips:**
  - EC2 = **full control compute**
  - Spot Instances = **cost optimization**

**AWS Lambda**

- **What it is:**
  - Serverless compute service to run code without servers
- **Core Features:**
  - Event-driven execution
  - Auto scaling
  - Pay per execution
  - Integrates with AWS services
- **When to Use:**
  - APIs (with API Gateway)
  - Automation tasks
  - Event processing
- **Not Ideal For:**
  - Long-running jobs (>15 minutes)
- **Exam Tips:**
  - Lambda = **serverless + event-driven**
  - Max timeout = 15 minutes

**AWS Serverless Application Model**

- **What it is:**
  - Framework to build and deploy **serverless applications**
- **Core Features:**
  - Simplified CloudFormation templates
  - Local testing support
  - Easy deployment of Lambda-based apps

- **When to Use:**
  - Serverless application development
  - Infrastructure as code for serverless
- **Not Ideal For:**
  - Non-serverless architectures
- **Exam Tips:**
  - SAM = **serverless IaC + deployment framework**

## Containers

**Amazon Elastic Container Registry**

- **What it is:**
  - Managed service to store Docker container images
- **Core Features:**
  - Secure image storage
  - IAM integration
  - Image versioning
- **When to Use:**
  - Store images for ECS/EKS
- **Not Ideal For:**
  - Running containers
- **Exam Tips:**
  - ECR = **Docker image repository**

**Amazon Elastic Container Service**

- **What it is:**
  - AWS-native service to run and manage containers
- **Core Features:**
  - Launch types: EC2 & Fargate
  - Tight AWS integration

- Load balancing support

• **When to Use:**

- Microservices architecture
- Containerized apps

• **Not Ideal For:**

- Kubernetes-based workloads

• **Exam Tips:**

- ECS = **simple container orchestration**
- Fargate = **serverless containers**

**Amazon Elastic Kubernetes Service**

• **What it is:**

- Managed Kubernetes service

• **Core Features:**

- Managed control plane
- Supports EC2 & Fargate
- Highly scalable

• **When to Use:**

- Kubernetes-based applications
- Multi-cloud/container portability

• **Not Ideal For:**

- Simple container workloads

• **Exam Tips:**

- EKS = **Kubernetes on AWS**

## Database

**Amazon DocumentDB**

• **What it is:**

- Managed **document database** compatible with MongoDB

- **Core Features:**
  - JSON-like document storage
  - High availability (replication)
  - Managed backups and scaling
- **When to Use:**
  - MongoDB workloads
  - JSON/document-based applications
- **Not Ideal For:**
  - Relational queries
  - Complex joins
- **Exam Tips:**
  - DocumentDB = **MongoDB-compatible service**

**Amazon DynamoDB**

- **What it is:**
  - Fully managed **NoSQL key-value and document database**
- **Core Features:**
  - Single-digit millisecond latency
  - Auto scaling
  - Global tables
  - Streams + TTL
- **When to Use:**
  - Serverless applications
  - High-scale workloads
- **Not Ideal For:**
  - Complex joins
- **Exam Tips:**
  - DynamoDB = **serverless NoSQL DB**
  - Partition key design is critical

**Amazon Keyspaces**

- **What it is:**
  - Managed **Apache Cassandra-compatible database**
- **Core Features:**
  - Serverless
  - High availability
  - Compatible with Cassandra Query Language (CQL)
- **When to Use:**
  - Cassandra workloads
  - High write throughput systems
- **Not Ideal For:**
  - Relational workloads
- **Exam Tips:**
  - Keyspaces = **Cassandra on AWS**

**Amazon MemoryDB for Redis**

- **What it is:**
  - Redis-compatible database with **durable storage**
- **Core Features:**
  - In-memory performance
  - Persistent storage
  - High availability
- **When to Use:**
  - Ultra-fast applications with durability
  - Real-time apps
- **Not Ideal For:**
  - Complex queries
- **Exam Tips:**

- MemoryDB = **Redis + durability (DB, not just cache)**

**Amazon Neptune**

• **What it is:**

- Fully managed **graph database**

• **Core Features:**

- Supports Gremlin and SPARQL
- High-performance graph queries
- Highly available

• **When to Use:**

- Social networks
- Recommendation engines
- Fraud detection

• **Not Ideal For:**

- Relational or key-value use cases

• **Exam Tips:**

- Neptune = **graph database (relationships focus)**

**Amazon RDS**

• **What it is:**

- Managed **relational database service**

• **Core Features:**

- Supports MySQL, PostgreSQL, SQL Server, Oracle
- Multi-AZ for high availability
- Read replicas for scaling

• **When to Use:**

- Traditional relational applications

• **Not Ideal For:**

- Massive NoSQL workloads

- **Exam Tips:**
  - Multi-AZ = **high availability**
  - Read replicas = **read scaling**

**Amazon Aurora**

- **What it is:**
  - High-performance relational database compatible with MySQL/PostgreSQL
- **Core Features:**
  - Up to 5x faster than MySQL
  - Auto-scaling storage
  - Multi-AZ by default
- **When to Use:**
  - High-performance relational workloads
- **Not Ideal For:**
  - NoSQL workloads
- **Exam Tips:**
  - Aurora = **high-performance RDS alternative**

**Amazon Redshift**

- **What it is:**
  - Fully managed **data warehouse** for analytics
- **Core Features:**
  - Columnar storage
  - Massively parallel processing (MPP)
  - Integrates with S3 and BI tools
- **When to Use:**
  - Data warehousing
  - Analytics and reporting
- **Not Ideal For:**

- OLTP workloads

• **Exam Tips:**

- Redshift = **analytics/data warehouse**
- Often paired with **S3 + BI tools**

## Developer Tools

**AWS Command Line Interface**

• **What it is:**

- Command-line tool to interact with AWS services

• **Core Features:**

- Manage AWS resources via terminal
- Supports scripting and automation
- Works with profiles and credentials

• **When to Use:**

- Automation tasks
- Managing AWS from terminal

• **Not Ideal For:**

- GUI-based management

• **Exam Tips:**

- CLI = **automation + scripting**

**AWS CloudFormation**

• **What it is:**

- Infrastructure as Code using JSON/YAML templates

• **Core Features:**

- Stack-based deployment
- Dependency management
- Automated provisioning

• **When to Use:**

- Repeatable infrastructure setup

• **Exam Tips:**

- CloudFormation = **template-based IaC**

**AWS Cloud Development Kit**

• **What it is:**

- Define infrastructure using programming languages

• **Core Features:**

- Supports Python, TypeScript, Java
- Converts code → CloudFormation
- Reusable constructs

• **When to Use:**

- Developer-friendly IaC

• **Exam Tips:**

- CDK = **code-based IaC**

**AWS CodeBuild**

• **What it is:**

- Managed service to build and test code

• **Core Features:**

- Compile, test, package code
- Serverless (no infra)
- Integrates with CodePipeline

• **When to Use:**

- CI pipelines

• **Exam Tips:**

- CodeBuild = **build stage**

**AWS CodeDeploy**

- **What it is:**
  - Automated deployment service
- **Core Features:**
  - Blue/Green deployments
  - Rolling updates
  - Supports EC2, Lambda
- **When to Use:**
  - Application deployments
- **Exam Tips:**
  - CodeDeploy = **deployment strategies**

**AWS CodePipeline**

- **What it is:**
  - CI/CD pipeline orchestration service
- **Core Features:**
  - Automates build, test, deploy stages
  - Integrates with CodeBuild, CodeDeploy
- **When to Use:**
  - End-to-end CI/CD pipelines
- **Exam Tips:**
  - CodePipeline = **pipeline automation**

**Amazon Q**

- **What it is:**
  - AI-powered assistant for developers
- **Core Features:**
  - Code suggestions
  - AWS best practices guidance
  - Debugging assistance

- **When to Use:**
  - Improve developer productivity
- **Exam Tips:**
  - Amazon Q = **AI coding assistant**

## Web & Mobile

**Amazon API Gateway**

- **What it is:**
  - Managed service to create and manage APIs
- **Core Features:**
  - REST and WebSocket APIs
  - Throttling and caching
  - Authentication (IAM, Cognito)
- **When to Use:**
  - Exposing backend services
  - Serverless APIs
- **Not Ideal For:**
  - Internal-only services (use ALB)
- **Exam Tips:**
  - API Gateway = **API + Lambda integration**

## Machine Learning

**Amazon SageMaker AI**

- **What it is:**
  - Fully managed platform for building, training, and deploying ML models
- **Core Features:**
  - Managed notebooks
  - Model training and tuning

- Real-time and batch inference
- AutoML

• **When to Use:**

- Machine learning workflows
- AI applications

• **Exam Tips:**

- SageMaker = **end-to-end ML lifecycle**

**Amazon Bedrock**

• **What it is:**

- Service to build **generative AI applications using foundation models**

• **Core Features:**

- Access to foundation models (Anthropic, AI21, etc.)
- No need to manage infrastructure
- Supports prompt engineering

• **When to Use:**

- Generative AI apps
- Chatbots, content generation

• **Exam Tips:**

- Bedrock = **Generative AI without managing models**

**Amazon Kendra**

• **What it is:**

- Intelligent search service powered by ML

• **Core Features:**

- Natural language search
- Connects multiple data sources
- ML-powered ranking

• **When to Use:**

- Enterprise search
- Knowledge base search

• **Exam Tips:**

- Kendra = **intelligent search (NLP-based)**

## Management & Governance

**AWS CloudTrail**

• **What it is:**

- Service to track **API calls and user activity** in AWS

• **Core Features:**

- Logs all API activity
- Stores logs in S3
- Supports auditing and compliance

• **When to Use:**

- Security monitoring
- Compliance tracking

• **Exam Tips :**

- CloudTrail = **who did what in AWS**

**Amazon CloudWatch**

• **What it is:**

- Monitoring service for **metrics, logs, and alarms**

• **Core Features:**

- Metrics dashboards
- Alerts and alarms
- Integration with AWS services

• **When to Use:**

- Performance monitoring
- System health tracking

- **Exam Tips :**
  - CloudWatch = **monitoring + alerts**

**Amazon CloudWatch Logs**

- **What it is:**
  - Service to **collect, store, and analyze logs**
- **Core Features:**
  - Log groups and streams
  - Real-time log monitoring
  - Integration with Lambda, EC2
- **When to Use:**
  - Application logging
  - Debugging
- **Exam Tips :**
  - CloudWatch Logs = **log storage + analysis**

**AWS Config**

- **What it is:**
  - Tracks **resource configuration and compliance**
- **Core Features:**
  - Configuration history
  - Compliance rules
  - Drift detection
- **When to Use:**
  - Compliance auditing
  - Configuration tracking
- **Exam Tips :**
  - Config = **resource tracking + compliance**

**Amazon Managed Grafana**

• **What it is:**

- Managed service for **Grafana dashboards**

• **Core Features:**

- Visualization dashboards
- Integrates with CloudWatch, Prometheus

• **When to Use:**

- Monitoring dashboards

• **Exam Tips :**

- Grafana = **visual monitoring dashboards**

**AWS Systems Manager**

• **What it is:**

- Manage and automate **EC2 and infrastructure operations**

• **Core Features:**

- Run Command
- Patch Manager
- Parameter Store

• **When to Use:**

- Remote EC2 management
- Automation

• **Exam Tips :**

- Systems Manager = **manage EC2 without SSH**

**AWS Well-Architected Tool**

• **What it is:**

- Tool to review workloads against **AWS best practices**

• **Core Features:**

- Evaluates architecture

- Covers pillars (security, cost, performance, etc.)

• **When to Use:**

- Architecture review
- Improving system design

• **Exam Tips :**

- Well-Architected Tool = **best practice evaluation**

**AWS Data Exchange**

• **What it is:**

- Service to **find, subscribe, and use third-party datasets**

• **Core Features:**

- Data marketplace
- Secure data sharing
- Integration with S3

• **When to Use:**

- Access external datasets
- Data monetization

• **Exam Tips :**

- Data Exchange = **external data marketplace**

## Migration & Transfer

**AWS Application Discovery Service**

• **What it is:**

- Collects information about on-premises applications

• **Core Features:**

- Server inventory
- Dependency mapping
- Migration planning

• **When to Use:**

- Planning migration to AWS

• **Exam Tips :**

- Discovery Service = **understand current environment**

**AWS Application Migration Service**

• **What it is:**

- Service to **migrate applications to AWS**

• **Core Features:**

- Lift-and-shift migration
- Continuous replication
- Minimal downtime

• **When to Use:**

- Migrating servers to AWS

• **Exam Tips :**

- MGN = **lift-and-shift migration**

**AWS Database Migration Service**

• **What it is:**

- Service to migrate databases to AWS

• **Core Features:**

- Supports homogeneous & heterogeneous migrations
- Continuous data replication
- Minimal downtime

• **When to Use:**

- Database migration

• **Exam Tips :**

- DMS = **database migration**

**AWS DataSync**

- **What it is:**
  - Service to transfer large amounts of data
- **Core Features:**
  - Fast data transfer
  - Automated replication
  - Supports on-prem → AWS
- **When to Use:**
  - Large data transfers
- **Exam Tips :**
  - DataSync = **fast data transfer**

**AWS Snow Family**

- **What it is:**
  - Physical devices to transfer large data to AWS
- **Core Features:**
  - Offline data transfer
  - Snowball, Snowcone, Snowmobile
  - Secure and durable
- **When to Use:**
  - TB–PB scale data transfer
  - Limited network connectivity
- **Exam Tips :**
  - Snow Family = **offline data transfer**

**AWS Transfer Family**

- **What it is:**
  - Managed service for file transfers using FTP, SFTP, FTPS
- **Core Features:**
  - Secure file transfer

- Integrates with S3/EFS
- Supports legacy systems

• **When to Use:**

- File transfer workflows
- Legacy integration

• **Exam Tips :**

- Transfer Family = **FTP/SFTP on AWS**

## Networking & Content Delivery

**Amazon CloudFront**

• **What it is:**

- Content Delivery Network (CDN) to deliver content with low latency

• **Core Features:**

- Global edge locations
- Caching of static/dynamic content
- Integration with S3, ALB, EC2

• **When to Use:**

- Deliver content globally
- Improve website performance

• **Not Ideal For:**

- Internal/private-only access (use VPC/PrivateLink)

• **Exam Tips:**

- CloudFront = **low latency + global CDN**

**AWS PrivateLink**

• **What it is:**

- Enables **private connectivity between VPCs and AWS services**

• **Core Features:**

- No internet gateway required

- Secure private access
- Uses VPC endpoints

• **When to Use:**

- Private communication between services
- Secure access to AWS services

• **Not Ideal For:**

- Public internet access

• **Exam Tips:**

- PrivateLink = **private access without internet**

**Amazon Route 53**

• **What it is:**

- Managed DNS and domain routing service

• **Core Features:**

- Domain registration
- Routing policies (latency, failover, weighted)
- Health checks

• **When to Use:**

- DNS management
- Traffic routing

• **Exam Tips:**

- Route 53 = **DNS + routing policies**

**Amazon VPC**

• **What it is:**

- Virtual private network in AWS

• **Core Features:**

- Subnets (public/private)
- Internet Gateway, NAT Gateway

- Security Groups and NACLs

• **When to Use:**

- Secure network setup
- Isolated environments

• **Exam Tips:**

- VPC = **network isolation + control**

## Security, Identity & Compliance

**AWS Identity and Access Management**

• **What it is:**

- Service to manage **users, roles, and permissions**

• **Core Features:**

- Policies (JSON-based)
- Role-based access
- Fine-grained permissions

• **When to Use:**

- Controlling access to AWS resources

• **Not Ideal For:**

- End-user authentication (use Cognito)

• **Exam Tips:**

- IAM = **access control**
- Follow **least privilege principle**

**AWS Key Management Service**

• **What it is:**

- Managed service to create and manage encryption keys

• **Core Features:**

- Key creation and rotation
- Integration with AWS services

- Secure key storage

• **When to Use:**

- Encrypting data at rest

• **Exam Tips:**

- KMS = **encryption key management**

**Amazon Macie**

• **What it is:**

- Service to discover and protect **sensitive data in S3**

• **Core Features:**

- Detects PII (personal data)
- Data classification
- Alerts on sensitive data exposure

• **When to Use:**

- Data security and compliance

• **Exam Tips:**

- Macie = **PII detection in S3**

**AWS Secrets Manager**

• **What it is:**

- Secure storage for secrets like passwords and API keys

• **Core Features:**

- Automatic rotation
- Fine-grained access control
- Integration with RDS, Lambda

• **When to Use:**

- Managing sensitive credentials

• **Not Ideal For:**

- General config (use Parameter Store)

• **Exam Tips:**

- Secrets Manager = **store & rotate secrets**

**AWS Shield**

• **What it is:**

- Managed service for **DDoS protection**

• **Core Features:**

- Standard (free) and Advanced versions
- Protects against network attacks
- Integrated with CloudFront, Route 53

• **When to Use:**

- Protect applications from DDoS attacks

• **Exam Tips:**

- Shield = **DDoS protection**

**AWS WAF**

• **What it is:**

- Web Application Firewall to protect applications

• **Core Features:**

- Blocks SQL injection, XSS
- Rule-based filtering
- Integrates with CloudFront and ALB

• **When to Use:**

- Protect web applications

• **Exam Tips:**

- WAF = **application-layer protection**

## Storage

**AWS Backup**

- **What it is:**
  - Centralized service to **automate backups across AWS services**
- **Core Features:**
  - Backup policies and scheduling
  - Supports EBS, RDS, DynamoDB, EFS, etc.
  - Centralized backup management
  - Lifecycle management (retention rules)
- **When to Use:**
  - Automating backups
  - Compliance and data protection
- **Not Ideal For:**
  - Real-time data access
- **Exam Tips :**
  - AWS Backup = **centralized backup management across services**

**Amazon Elastic Block Store**

- **What it is:**
  - Block storage for **EC2 instances**
- **Core Features:**
  - Persistent storage
  - High performance (SSD/HDD options)
  - Snapshots for backup
- **When to Use:**
  - OS disks
  - Databases on EC2
- **Not Ideal For:**
  - Shared storage across multiple instances
- **Exam Tips :**
  - EBS = **block storage attached to EC2**

**Amazon Elastic File System**

- **What it is:**
  - Managed **file storage system (NFS)**
- **Core Features:**
  - Shared across multiple EC2 instances
  - Auto scaling
  - Highly available
- **When to Use:**
  - Shared file systems
  - Linux workloads
- **Not Ideal For:**
  - Windows workloads
- **Exam Tips :**
  - EFS = **shared storage for multiple EC2s**

**Amazon S3**

- **What it is:**
  - Scalable **object storage service**
- **Core Features:**
  - Unlimited storage
  - Versioning
  - Lifecycle policies
  - Event notifications
- **When to Use:**
  - Static files
  - Backups
  - Data lakes
- **Exam Tips :**

- S3 = **object storage**
- Storage classes (Standard, IA, Glacier)

**Amazon S3 Tables**

• **What it is:**

- Feature to store and manage **tabular datasets in S3**

• **Core Features:**

- Optimized for analytics workloads
- Supports structured/tabular data
- Integration with analytics services

• **When to Use:**

- Data lakes with structured datasets
- Analytics pipelines

• **Not Ideal For:**

- Unstructured object storage

• **Exam Tips :**

- S3 Tables = **structured data in S3 (analytics focus)**

**Amazon S3 Glacier**

• **What it is:**

- Low-cost storage for **archival data**

• **Core Features:**

- Very low cost
- Retrieval times (minutes to hours)
- Long-term storage

• **When to Use:**

- Archival and backup data
- Compliance storage

• **Not Ideal For:**

- Frequent access

- **Exam Tips:**
  - Glacier = **cheap storage, slow retrieval**